MILITARY SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT LOW LEVEL TO 2.0 AMPERE (LATCHING) WITH INTERNAL DIODES FOR COIL TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-39016

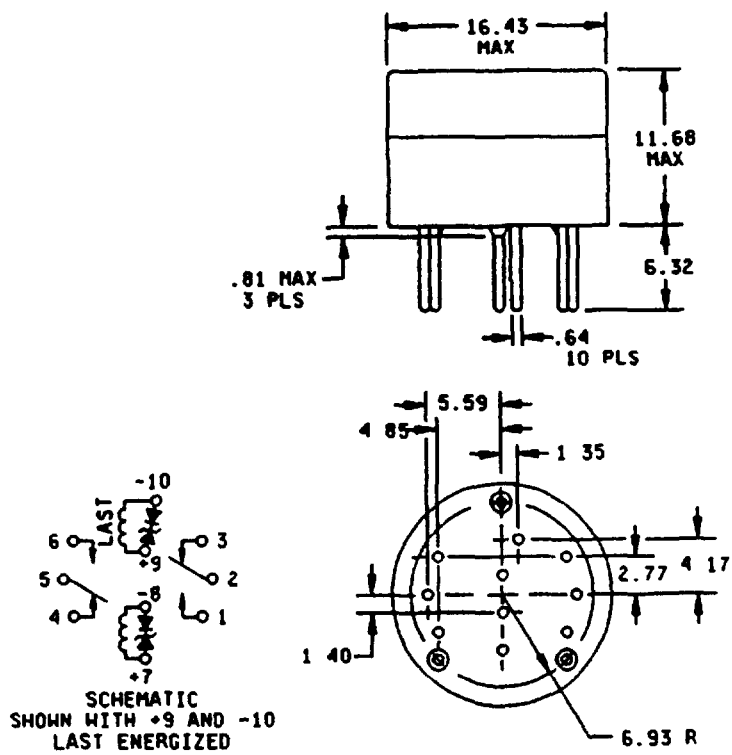


FIGURE 1. Dimensions and configuration

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mm	Inches	mm	Inches
0.64	.025	4.85	.191
0.81	.032	5.59	.220
1.35	.053	6.32	.249
1.40	.055	6.93	.273
2.77	.109	11.68	.460
4.17	.164	16.43	.647

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±0.25 mm (.010 inch).
4. All leads shall be electrically insulated from the case.
5. Circuit diagram shown on part is the terminal view.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Socket pin terminals shall be gold plated in accordance with MIL-G-45204, type II, class 1, over underplating nickel 1.25 to 3.75 µm (50 x 10^{-6} to 150 x 10^{-6} inch) thick.
8. Solder pin terminals shall be hot solder dipped.
9. The US Government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:
2.0 ampere at 28 V dc.
250 mA at 115 V ac 60 and 400 Hz case not grounded.
100 mA at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.125 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.225 ohm maximum.

Low level:

During life: 33 ohms maximum.

After life: 0.175 ohm maximum.

Intermediate current:

During life: 1 ohm maximum.

After life: 0.225 ohm maximum.

Contact bounce: 2.2 ms maximum (applicable to failure rate level "L").

Contact stabilization time: 2.5 ms maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated resistive current. Not applicable to ac load ratings.

Neutral screen: Applicable.

COIL DATA: See table I.

Operate time: 4.5 ms maximum over temperature range with rated coil voltage.

Release time: Not applicable.

ELECTRICAL DATA:

Insulation resistance: 10,000 megohms at 500 V dc.

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Dielectric withstanding voltage:

Sea level:

Between case, frame or enclosure and all contacts both in the energized and deenergized positions	500 V rms (60 Hz)
Between case, frame, or enclosure and coils	500 V rms (60 Hz)
Between all contacts and coils	500 V rms (60 Hz)
Between open contacts in the energized and deenergized positions	500 V rms (60 Hz)
Between open contacts	500 V rms (60 Hz)

Altitude:

Between all terminals to case	125 V rms (60 Hz)

DIODE CHARACTERISTICS:

Coil transient suppression: Applicable.

Diode block integrity (delete coil resistance and perform this test after coil transient suppression test in all inspection tables of MIL-R-39016): With applicable voltage applied to the relay coil circuit in the reverse direction, monitor leakage current with dc micrometer, oscilloscope, or qualifying activity approved test equipment. Leakage current shall not exceed the specified value.

Block integrity maximum leakage current: 1 μA at 50 V dc.

Maximum back emf: See table I.

Breakdown voltage: 100 V dc minimum at 10 μA. (This test may be performed in-process or as final assembly).

Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750 methods 2073 or 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C

Vibration (sinusoidal) 1/: MIL-STD-202, method 204, test condition G. Contact chatter shall not exceed 10 μs maximum for closed contacts, and 1 μs maximum closure for open contacts.

Vibration (random) 1/: MIL-STD-202, method 214, test condition HG. Contact chatter shall not exceed 10 μs maximum for closed contacts, and 1 μs maximum closure for open contacts. Applicable to qualification and group C.

Shock (specified pulse) 1/: MIL-STD-202, method 213, except 3,500 g's half-sine, 2.0 ms. Contact chatter shall not exceed 10 μs maximum for closed contacts, and 1 μs maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with method 1041 of MIL-STD-750.

1/ Relays under test are hard wired and potted in wax.

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Moisture resistance: Moisture resistance test shall be performed in accordance with method 106 of MIL-STD-202 immediately following terminal strength in group III of qualification inspection and group II of group C inspection. The following details apply:

Initial measurements: Not applicable.
Polarization: Applicable.
Loading voltage: Not applicable.
Final measurements: At high humidity and after high humidity not applicable. After drying period - Insulation resistance, dielectric withstanding voltage, static contact resistance and latch/reset voltage shall be within specification limits.

PHYSICAL DATA:

Terminal strength, method 211 of MIL-STD-202.

Pull test: Test condition A, 2.2 kg (1 pound) pull.
Bend test: Test condition C, 1.1 kg (0.5 pound) pull.
Twist test: As specified in MIL-R-39016.

Solderability: Applicable.

Dimensions and configuration: See figure 1.

Design and construction: The contacts are contained in a hermetically sealed chamber which does not contain the coil. The coils are located outside the hermetically sealed chamber and protected by encapsulation.

Weight: 8.0 grams (0.282 ounce) maximum.

Seal: Hermetic, (less than 1×10^{-9} cm^3/s Helium at contact chamber). Seal test shall be 100 percent inspection and shall be performed as an in-process test on the contact chamber only. Seal testing is not applicable to qualification and quality conformance inspections.

Minimum marking: Military Part or Identifying Number (PIN), "J" with the date code (example J9237), circuit diagram, manufacturers' name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.
Low level: 100,000 cycles plus 900,000 cycles mechanical life.

PIN: M39016/59- (plus dash number from table I and suffix letter designation failure rate level as described in footnote 2/ of table I).

TABLE I. Dash numbers and characteristics. 1/

Dash number 2/		Coil voltage (V dc) 3/		At +25°C 4/		Over temp. range 4/	At +25°C
Socket pin	Solder pin	Rated	Max	Coil resistance ohms ±10% (@ +25°C) 5/	Latch/Reset voltage (maximum) (V dc)	Latch/Reset voltage (maximum @ +125°C) (V dc)	Max. back emf (V dc)
001	002	5.0	5.8	28	2.8	3.5	9.0
003	004	6.0	7.3	40	3.0	4.4	9.0
005	006	12.0	14.8	160	6.0	8.4	18.0
007	008	26.5	32.0	750	13.0	18.0	42.0

See footnotes at end of table.

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TABLE I Dash numbers and characteristics - Continued 1/

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number Failure rate level (percent per 10,000 cycles) L, 3.0, M, 1.0; P, 0.1, R, 0.01 Example: -004L

3/ CAUTION The use any coil voltage less than the rated voltage will compromise the operation of the relay

4/ Latch/Reset values as shown are for test purposes only and are not to be used as design criteria

5/ Coil resistance is not directly measurable at relay terminals When rated voltage is applied to coil terminals, the coil circuit must be within the limits shown. Measure at 25°C at nominal voltage for 5 seconds maximum

QUALIFICATION INSPECTION

Qualification inspection and sample size See table II.

TABLE II. Qualification inspection and sample size

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 1/ 33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable	M39016/59-007	18 units plus 1 open unit for level L at C = 0 1/ 33 units plus 1 open unit for level M at C = 0 1/ Qualification inspection as applicable
	M39016/59-001 M39016/59-003 M39016/59-005 M39016/59-008	2 units each PIN qualification inspection table, group II 1 unit terminal strength and solderability

1/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the contractor elects to test the number of units permitting one or more failures Prior to performance of qualification inspection, the relay manufacturer shall preselect the sampling plan

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Preparing activity.
Navy - EC

Agent
DLA - ES

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